CUBICLE NO 8

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Day 12 – Buzzer Interfacing with Arduino Uno

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1. Introduction to Buzzer

A **buzzer** is an audio signaling device used to produce sound.
It can be:

- **Active buzzer** (produces sound when powered — no need for tone generation)
- **Passive buzzer** (requires specific frequency signal to generate sound)

Common uses:

- Alarms and notifications
- Timers and indicators
- Sound effects in electronics

2. Pin Description (KY-012 Buzzer Module)

Pin	Description
S	Signal (connect to Arduino digital pin)
+	VCC (5V)
-	GND

3. Requirements

- Arduino Uno
- KY-012 buzzer module (or passive buzzer)

- Jumper wires
- Optional LED (for visual indication)

4. Working Principle

- **Active buzzer**: Just apply HIGH signal → sound is produced
- **Passive buzzer**: Use Arduino's tone() function to send a frequency signal
- Sound frequency changes with tone() parameter (in Hz)

5. Circuit Diagram (Basic Connection)

- **Buzzer S → Arduino D8 or D9**
- **Buzzer + → Arduino 5V**
- **Buzzer - → Arduino GND**
- If LED used: Connect LED anode to Arduino digital pin via 220Ω resistor, cathode to GND

6. Code Examples and Explanations

Example 1 – Basic ON/OFF Buzzer

```
int buzzer = 9;  // Pin connected to KY-012 signal pin

void setup() {
  pinMode(buzzer, OUTPUT);
}

void loop() {
```

```
  digitalWrite(buzzer, HIGH);  // Turn buzzer ON
  delay(1000);                 // Wait 1 second
  digitalWrite(buzzer, LOW);   // Turn buzzer OFF
  delay(1000);                 // Wait 1 second
}
```

Explanation:

- Turns buzzer ON for 1 second, OFF for 1 second (continuous beeping)
- Works with **active buzzer** only

Output:

- Continuous beeping with 1-second interval

Example 2 – Buzzer + LED Blink

```
#define buzzerPin 8
#define ledPin 7

void setup() {
  pinMode(buzzerPin, OUTPUT);
  pinMode(ledPin, OUTPUT);
}

void loop() {
  digitalWrite(buzzerPin, HIGH);
  digitalWrite(ledPin, HIGH);
  delay(500);
```

```
  digitalWrite(buzzerPin, LOW);
  digitalWrite(ledPin, LOW);
  delay(500);
}
```

Explanation:

- Buzzer and LED turn ON/OFF together
- ON for 0.5s, OFF for 0.5s

Output:

- Beep and LED flash simultaneously

Example 3 – Passive Buzzer with Tone

```
#define buzzerPin 8
#define ledPin 7

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  tone(buzzerPin, 1000);  // 1 kHz tone
  digitalWrite(ledPin, HIGH);
  delay(300);

  noTone(buzzerPin);      // Stop sound
  digitalWrite(ledPin, LOW);
```

```
  delay(300);
}
```

Explanation:

- Uses tone() to produce a **1000 Hz sound**
- LED lights up while buzzer is active

Output:

- High-pitched beep with LED blink every 0.3s

Example 4 – Frequency Sweep

```
#define buzzerPin 8
#define ledPin 7

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  for (int i = 1000; i <= 2000; i += 10) {
    tone(buzzerPin, i);
    digitalWrite(ledPin, !digitalRead(ledPin)); // Toggle LED
    delay(5);
  }

  for (int i = 2000; i >= 1000; i -= 10) {
    tone(buzzerPin, i);
```

```
    digitalWrite(ledPin, !digitalRead(ledPin));
    delay(5);
  }
}
```

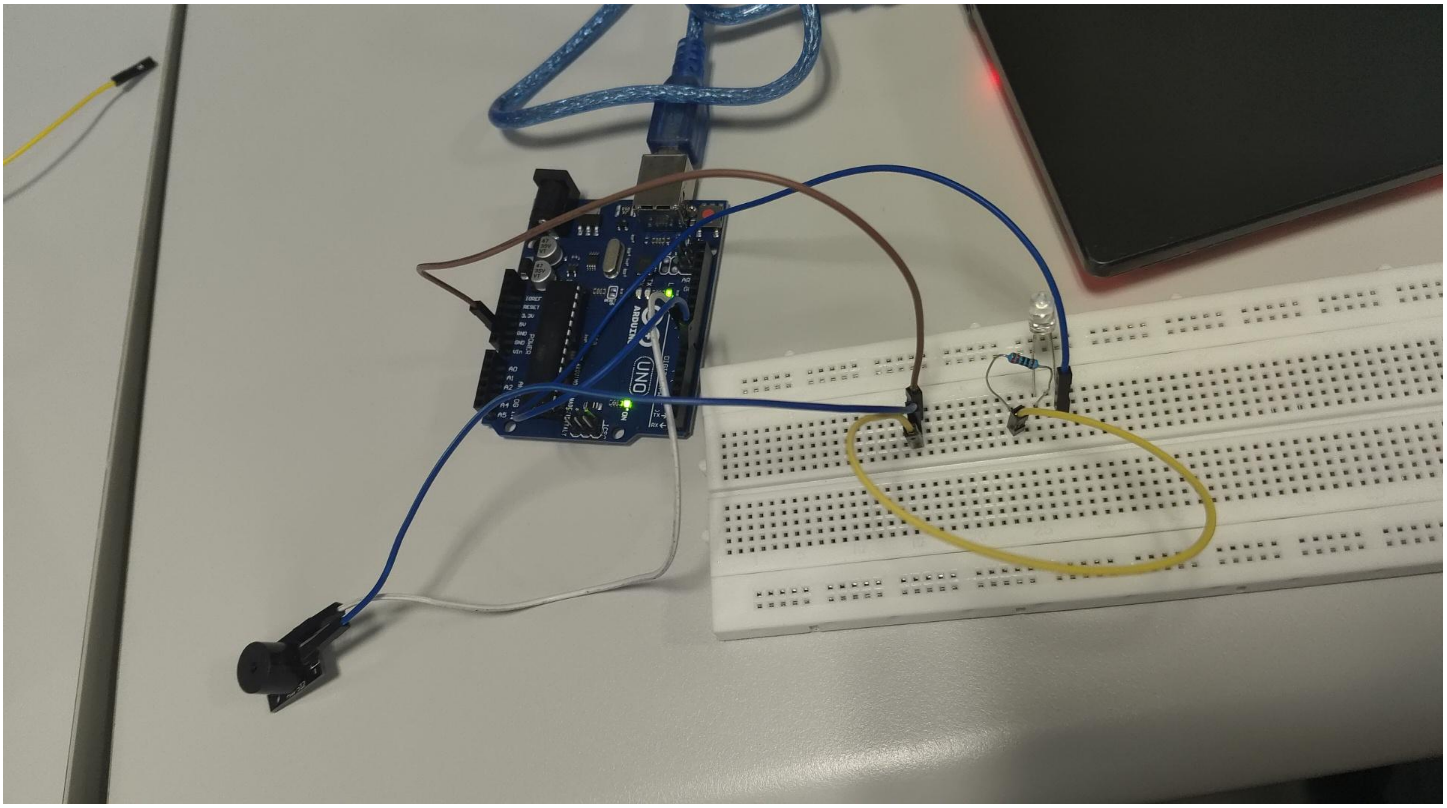
Explanation:

- Gradually increases pitch from **1000 Hz to 2000 Hz** and back down
- LED toggles at each frequency change

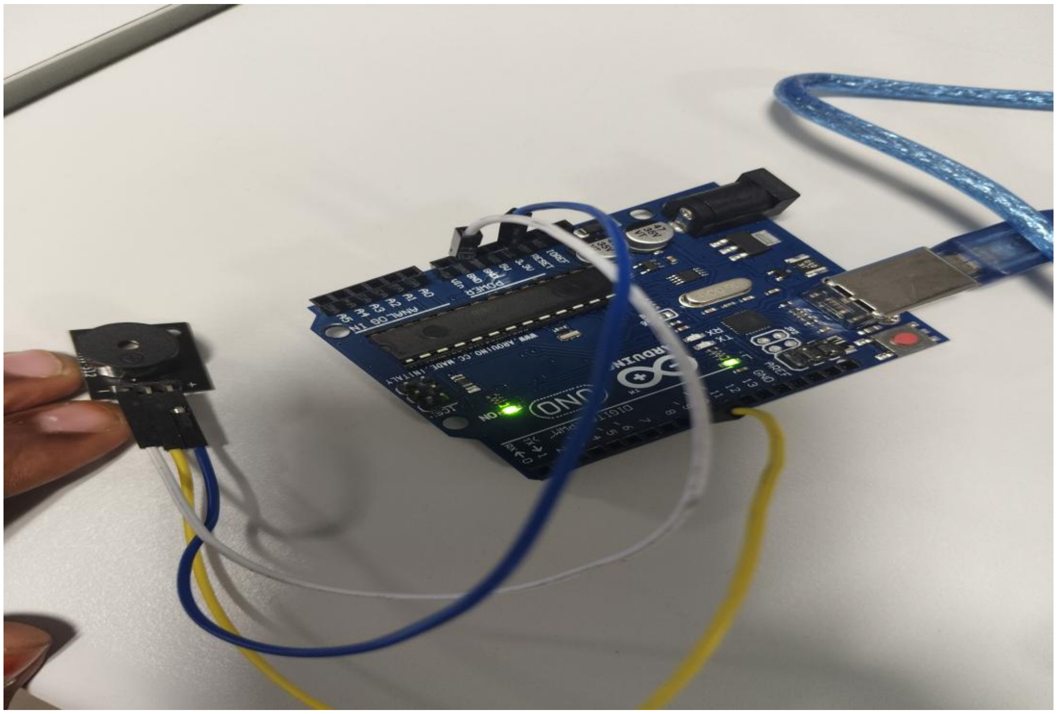
Output:

- Rising and falling siren-like sound with LED flashing rapidly

OUPUT IMAGES AND VIDEOS:



108.mp4 ECG.mp4 REVERSE.mp4



7. Advantages

- Simple to use
- Low cost
- Low power consumption
- Works with Arduino easily

8. Disadvantages

- Limited volume (for small buzzers)
- Not suitable for music playback (unless using tone() with precise frequencies)

9. Limitations

- Passive buzzers need PWM signal, not just HIGH/LOW
- Can only play one tone at a time with tone()

10. Applications

- Doorbell systems
- Alarms & alerts

- Timers and countdowns
- Game sound effects